Student Name: Student ID:

Habib University
Dhanani School of Science and Engineering
Computer Architecture
EE 371 / CS 330 / CE 321 (Fall 2025)

Quiz 3

Date: 7th Nov, 2025

Total marks: 20 **Obtained Marks:**

Total time: 50 mins

Instructions:

1. All questions should be answered **with ink pen.**
2. Smart watches, laptops, and similar electronics are **strictly NOT allowed**.
3. Answer sheets should contain all steps, working, explanations, and assumptions.
4. Write your name and HU ID on all sheets.
5. Turn in your question paper along with your answer sheets.
6. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This quiz assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	***Explain*** the role of ISA in modern processors and instruction encodings and assembly language programming	
CLO 2	***Explain*** the architecture and working of a single cycle processor	
CLO 3	***Design*** the architecture to mitigate issues of a pipelined processor	√
CLO 4	***Analyze** the* performance of cache operations	

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Q1: Let's say we have a 7-stage processor. Delay for each stage is given below in a tabular form.

Fetch	Decode 1	Decode 2	Execute 1	Execute 2	Memory	Write-back
250	250	300	250	300	300	300

a) What is the maximum speed-up we can achieve with the pipelined processor? **(1-point)**

Sol: 1950/300 = 6.5

b) Assume we have a program with 5 instructions. What is the speed-up for this program when it executes on pipelined version compared to single cycle/non-pipelined version? **(1-point)**

Sol: pipelined_cc = 7 + instr – 1 = 7 + 5 - 1 = 11

pipelined_time = 11*300 = 3300

non_pipelined_time = 5*1950 = 9750

speed_up = non_pipelined_time/ pipelined_time = 2.95

c) Assume we have another program with 500 instructions. What is the speed-up for this program when it executes on pipelined version compared to single cycle/non-pipelined version? **(1-point)**

Sol: pipelined_cc = 5 + instr – 1 = 7 + 500 - 1 = 506

pipelined_time = 506*300 = 151800

non_pipelined_time = 500*1950 = 975000

speed_up = non_pipelined_time/ pipelined_time = 6.42

d) Are the speed-ups in part b and c same? If yes, why? If not, why? **(1-point)**

Sol: No, they are different. We need enough/large number of instructions to see pipeline gain/throughput.

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Q2: Assume that the following sequence of instructions is executed on a five-stage pipelined RISC-V processor.

```
ld x29, 0(x10)
ld x30, 0(x29)
add x18, x29, x30
sub x14, x30, x18
and x10, x16, x17
```

a) If the processor has no forwarding or hazard detection unit, insert NOP(s) to ensure correct execution. Re-write the code sequence with NOP(s) inserted**. (3-points)**

Sol:

```
ld x29, 40(x10)
nop
nop
ld x30, 0(x29)
nop
nop
add x18, x29, x30
nop
nop
sub x14, x30, x18
and x10, x16, x17
```

b) Can we rearrange the original code to minimize the number of NOPs needed? If yes, re-organize and write the sequence of instructions. **(3-points)**

Sol:

```
ld x29, 40(x10)
and x10, x16, x17
nop
ld x30, 0(x29)
nop
nop
add x18, x29, x30
nop
nop
sub x14, x30, x18
```

c) Assume that the processor has both hazard detection unit and forwarding unit. Draw a multi-cycle diagram for the first ten cycles during the execution of **original code** and specify values of ForwardA, ForwardB, PCWrite, IF/IDWrite and Control MUX input for each cycle. PCWrite and IF/IDWrite are 0 when a hazard is detected, otherwise both signals are 1. Also, Control MUX value 0 means normal instruction, whereas 1 means hazard is detected. **(6-points)**

Instr	Cc0	Cc1	Cc2	Cc3	Cc4	Cc5	Cc6	Cc7	Cc8	Cc9
ld x29	F	D	E	M	W					
ld x30 NOP		F	D	-	-	-				
ld x30			F	D	E	M	W			
add NOP				F	D	-	-	-		
add					F	D	E	M	W	
sub						F	D	E	M	W
and							F	D	E	M
FwdA	-	-	00	00	01	00	00	00	00	-
FwdB	-	-	00	00	00	00	01	10	00	-
PCWrt	-	1	0	1	0	1	1	1	-	-
IF/IDWrt	-	1	0	1	0	1	1	1	-	-
Ctrl MUX	-	0	1	0	1	0	0	0	-	-

Q3: Suppose a loop executes 20 times, and the branch instruction is taken 19 times and not taken once (on the final iteration).
Compare the total number of miss-predictions for:

(a) **Always Taken**

(b) **Always Not Taken**

(c) **1-bit dynamic predictor** (assume not take at start)

(d) **2-bit dynamic predictor** (assume not take at start)

Explain your reasoning. **(4 points)**

Sol:

Always taken → 19 correct, 1 incorrect/miss-prediction

Always NOT taken → 1correct, 19 incorrect/miss-predictions

1-bit dynamic → 1st incorrect, 2-19 correct, 20th in-correct; 2 miss-predictions

2-bit dynamic → 1st and 2nd incorrect, 3-19 correct, 20th in-correct; 3 miss-predictions

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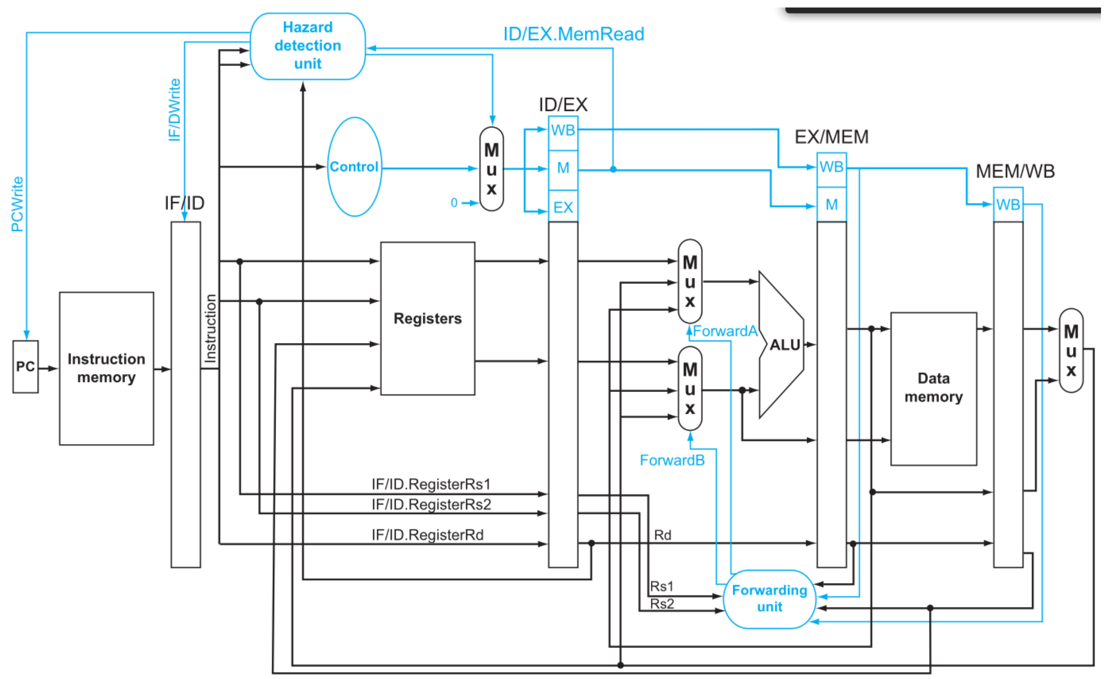

Mux control	Source	Explanation
ForwardA = 00	ID/EX	The first ALU operand comes from the register file.
ForwardA = 10	EX/MEM	The first ALU operand is forwarded from the prior ALU result.
ForwardA = 01	MEM/WB	The first ALU operand is forwarded from data memory or an earlier ALU result.
ForwardB = 00	ID/EX	The second ALU operand comes from the register file.
ForwardB = 10	EX/MEM	The second ALU operand is forwarded from the prior ALU result.
ForwardB = 01	MEM/WB	The second ALU operand is forwarded from data memory or an earlier ALU result.